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Original Article

Evolution toward monomorphy leads to a loss of morph plasticity in wing-polymorphic crickets

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Abstract

Threshold traits are widespread forms of dimorphism in which an underlying continuous liability produces discrete phenotypes once a threshold is crossed. The threshold trait model predicts that phenotypic variation will be observed only when the distribution of liability spans the threshold, and may otherwise remain latent, complicating predictions of how genetic and environmental variation shape responses to selection. We tested this prediction in the wing-polymorphic cricket *Gryllus lineaticeps*, in which wing length is a heritable, plastic threshold trait with an endocrine basis. We confirmed the final juvenile instar as the sensitive period for wing-morph determination and validated general esterase activity as a predictor of adult wing morph. We manipulated genetic variation in liability through directional selection on wing morph and then measured plasticity in wing morph and esterase activity. Selected populations, predicted to have extreme liability distributions not spanning the threshold, were non-plastic for wing-morph frequency despite maintaining plasticity in esterase activity. Thus, physiological plasticity persisted but became cryptic at the morphological level. These results support predictions of the threshold trait model and demonstrate that shifts in the liability distribution can mask environmental sensitivity, providing a framework for understanding the dynamic evolution and role of phenotypic plasticity in threshold traits.

Keywords: threshold traits, wing polymorphism, liability, phenotypic plasticity

Introduction

A variety of traits in natural populations display discontinuous variation, such that individuals tend to fall into distinct phenotypic forms rather than spanning a continuous range, resulting in distinct morphs that vary in morphology (e.g., male beetle horn length), behavior (e.g., guarding or sneaking male forms), or life history (e.g., insect-wing dimorphism) (Emlen, 2008; Roff, 1986, 1996). In some cases, these dimorphisms have a simple, Mendelian genetic basis, such as allelic variation at a single locus (Skuse, 2006; Yassin et al., 2016). However, many dimorphic traits can also be underlain by complex, multigenic architectures that are further shaped by environmental variation and potential gene-by-environment (G × E) interactions that shape trait plasticity (Reid & Acker, 2022; Roff, 1996). This disconnect between the genotype and discontinuous phenotype generates unique challenges for understanding the genetics, evolution, and plasticity governing threshold traits (Gianola, 1982; Lynch & Walsh, 1998; Reid & Acker, 2022; Roff, 1994, 1997, 1998; Treidel et al., 2026).

The threshold trait model provides a quantitative genetic framework for dimorphic traits with complex genetic bases (Figure 1A). In this model, an individual's liability—an unobserved, continuous underlying phenotype—reflects the combined effects of genetic variation, environmental conditions, and G × E interactions; and must exceed a critical threshold for a particular morph or trait to be expressed (Falconer, 1965; Roff, 1996). Alternate morphs are observed in a population when individuals' liability values in the population span the threshold. However, in populations where all individual liabilities are far from the threshold, genetic and environmental variation may be unobserved at the phenotypic level, leading to the apparent loss of the dimorphism (Figure 1B). This model provides a mechanism that may underlie evolutionary phenomena such as genetic assimilation, where changes in allele frequencies convert environmentally induced phenotypic variation to genetically encoded differences between individuals (Lande, 2009; Waddington, 1953) and may also explain patterns of trait gain and loss among closely related species (Reid & Acker, 2022; Treidel et al., 2026). However, there have been few empirical tests of this mechanism, in part because it remains challenging to link evolutionary change in threshold traits to the underlying physiological and developmental mechanisms (Zera & Denno, 1997; Zera & Harshman, 2001).

Wing-dimorphic insects present an excellent system for linking the physiological and developmental mechanisms of polymorphism to evolutionary change. Prevalent across

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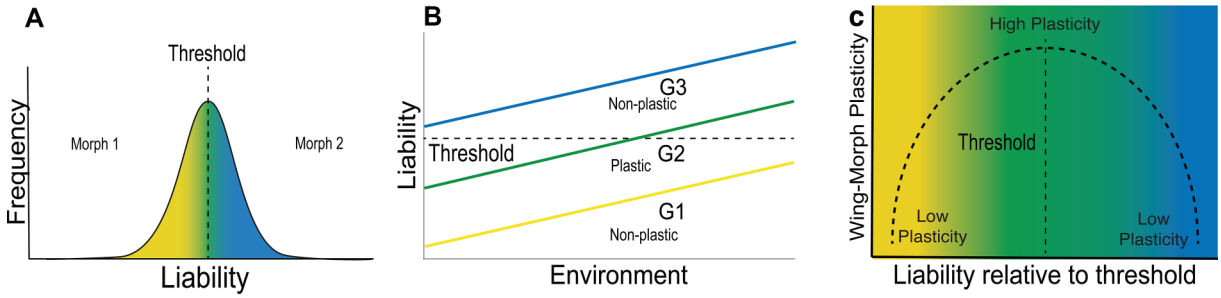


Figure 1. The threshold trait model describes a latent, normally distributed liability, jointly determined by genetic and environmental influences, that is transformed into a dimorphic trait by a developmental threshold. (A) Individuals with liabilities below the threshold develop into morph 1, whereas individuals with liabilities above the threshold develop into morph 2. (B) Phenotypic plasticity occurs when the distribution of liability overlaps the threshold. G1 and G3 represent liability distributions positioned primarily below or above the threshold, respectively, resulting in little or no observable plasticity. G2 represents a liability distribution overlapping the threshold, resulting in observable plasticity. (C) The threshold trait model predicts that plasticity is highest when liability distributions overlap the threshold and decreases when distributions are positioned away from the threshold.

many insect orders (Coleoptera, Diptera, Hymenoptera, Hemiptera, Orthoptera, and Homoptera), wing-dimorphic insects were foundational in the development of the threshold trait framework (Roff, 1986), and their study has generated a rich empirical foundation for genetic and endocrine mechanisms underlying polymorphisms (Nijhout, 1999; Zera, 2006; Zhang et al., 2019). Wing dimorphisms often exemplify a classic life-history tradeoff between dispersal and reproduction: Long-winged (LW) individuals invest in flight and dispersal capability, while short-winged (SW), or wingless individuals, forego flight to achieve earlier reproductive maturity (Zera & Denno, 1997). Insect wing dimorphisms have diverse modes of inheritance across taxa, including single-locus and polygenic architectures, sex-linked and autosomal genetic factors, and maternal effects (Braendle et al., 2006; Brisson, 2010; Roff, 1986). Wing morph can also be modulated by both abiotic and biotic factors (e.g., temperature, crowding, predation, host quality), and non-genetic parental effects (Hayes et al., 2024).

Among insects with wing-dimorphisms, crickets (genus *Gryllus*) provide a powerful system for investigating the evolutionary, developmental, and ecological mechanisms shaping the expression and evolution of wing dimorphism. Sixteen of the estimated 45 species in the North American *Gryllus* field cricket clade are wing dimorphic with multiple independent gains and losses of flight capability (Weissman & Gray, 2019). Quantitative genetic and artificial selection studies have demonstrated that wing polymorphism in four dimorphic *Gryllus* species is consistent with a polygenic mode of inheritance (Bégin & Roff, 2002; Crnokrak & Roff, 1998; Roff & Simons, 1997; Walker, 1987; Zera & Rankin, 1989). Also, less crowded, more nutritious, warmer, and longer daylength environments tend to increase the proportion of long-wing crickets in dimorphic species (Hayes et al., 2024; Roff et al., 2002; Treidel et al., 2021).

Hormone levels are hypothesized to be the physiological mechanism that converts liability to adult morph during *Gryllus* development (Fairbairn & Yadlowski, 1997; Zera & Denno, 1997; Zera & Holtmier, 1992; Zera & Tiebel, 1989). Juvenile hormone titer (JH) and activity of its degradation enzyme, juvenile hormone esterase (JHE), predict morph outcome in *Gryllus firmus*, *G. rubens*, and *G. assimilis* during the sensitive period for morph determination (Fairbairn & Yadlowski, 1997; Zera & Huang, 1999; Zera & Tiebel, 1989; Zera & Zhang, 1995). Both JH titers and JHE activity also change in response to selection on wing morph (Zera & Zhang, 1995). General esterase (GE) activity, reflecting the hydrolysis of ester bonds in substrates such as carboxylic esters, exhibits nearly identical developmental profiles to that of JHE in the last juvenile instar and is easily measured using a colorimetric assay (Gu & Zera, 1994, 1996; Woodring & Sparks, 1987). Thus, GE is a candidate indicator of physiological variation associated with the underlying liability that we can measure in individuals at the time of morph determination and before adult wing morph is expressed.

In this study, we established GE as a physiological indicator of the underlying liability during the sensitive period for wing development in the wing-polymorphic variable field cricket, *Gryllus lineaticeps*. We manipulated the underlying liability for morph determination using artificial selection populations and families derived from an outbred polymorphic *G. lineaticeps* colony, and reared crickets at different temperatures and day lengths. We found that populations predicted to have liabilities farther from the threshold remained plastic for underlying physiology but lost plasticity for the adult wing dimorphism, consistent with predictions from the threshold trait model (Figure 1B and C). Our results reveal how evolutionary shifts in underlying liability can alter, constrain, or eliminate phenotypic plasticity in a complex life-history trait while the underlying physiology remains sensitive to the environment and may explain dynamic patterns of trait reversal on phylogenies.

Materials and methods

Cricket stocks and husbandry

Breeding colonies of *G. lineaticeps* comprised at least 15 females and 8 males of each morph, placed on a mixture of soil and sand provided for oviposition. Egg cups were transferred to sealed containers with 50 ml conical tubes with water and plugged with cotton for hatching emergence. Hatchlings were housed under controlled densities to the penultimate juvenile stage (Treidel et al., 2021). Juvenile crickets were reared under a constant temperature and light cycle (26°C, 16h:8h L: D) with *ad libitum* access to water and a premixed and nutritionally complete dry lab diet including wheat germ, wheat bran, nutritional yeast, and milk powder (Zera & Larsen, 2001) and water.

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Polymorphic colony

The *G. lineaticeps* laboratory colony was established at the University of California, Berkeley, in July 2015, from more than 200 wild-collected *G. lineaticeps* from Sedgwick Reserve, University of California Natural Reserve System, UC Santa Barbara (34°41′34″N, 120°02′26″W, Santa Ynez, CA, USA). This colony serves as a genetically heterogeneous resource characterized by continuous, overlapping generations. We maintain morph ratios at 50:50 to ensure maintenance of the wing polymorphism and a range of liabilities. To maintain high levels of field-relevant genetic diversity, the colony is refreshed annually with the F1 offspring of at least 100 wild-caught individuals from the original source population at Sedgwick Reserve.

Artificial selection populations

To derive populations with liability distributions that were shifted away from the threshold, we initiated three replicate artificial selection populations from the genetically heterogeneous polymorphic colony. We established three separate founding populations (P1, P2, and P3) sampled from the polymorphic colony between August 2021 and March 2022. Each of these founding populations, which consisted of 44–60 each of LW and SW females, and 24–28 each of SW and LW males, was used to start a pair of selection populations: one selected for the SW morph and one for the LW morph, yielding three replicate pairs of lines (P1 LW, P1 SW, P2 LW, P2 SW, P3 LW, P3 SW).

Selection on adult wing phenotype was conducted in discrete, non-overlapping generations based on whether the hind wings extended past the abdomen. All crickets were at least 7-day-old adults when they entered a breeding colony to avoid inadvertent selection on early reproduction. For each subsequent generation, 78 females and 39 males of the selected morph were chosen to establish new breeding colonies. The experiments reported here were carried out on individuals from the eighth generation of selection (F8).

Identifying the sensitive period during development for wing morph plasticity

Previous work has demonstrated that morph frequency in the wing polymorphic field cricket *G. firmus* varies across seasons (Roff et al., 2002). Therefore, we manipulated photoperiod and temperature to test for seasonally relevant environmental effects on wing-morph determination in *G. lineaticeps* from the polymorphic colony. Prior work on the developmental endocrinology of wing morph identified differences in JHE activity between developing LW and SW individuals in the final juvenile stage (Gu & Zera, 1994; Zera & Huang, 1999; Zera & Tiebel, 1989). We therefore used individuals in the penultimate juvenile stage to capture developmental responses to environmental conditions before these endocrine differences emerge. Crickets at the penultimate instar (i.e., the second-to-last juvenile stage before adulthood) were identified by the presence of wing pads and placed into cohorts of roughly equal sex ratio (30 crickets per cohort, $n = 492$; 249 males, 243 females). Cohorts were randomly assigned to one of four treatments in temperature- and light-cycle-controlled incubators (I-36VL, Percival Scientific, IA, USA): (1) Constant 28°C, 16 h:8 h Light: Dark (Warm Long-Day, LD) ($n = 6$ cohorts); (2) Constant 24°C, 10 h:14 h Light: Dark (Cool Short-Day, SD) ($n = 6$ cohorts); (3) Transfer from Warm LD to Cool SD conditions ($n = 3$ cohorts); and (4) Transfer from Cool SD to Warm LD conditions ($n = 3$ cohorts). The transfer treatments were implemented when half of the individuals in a cohort had molted into the ultimate juvenile stage. All individuals were visually phenotyped for wing morph upon reaching adulthood to determine the morph frequency of the cohorts.

Establishing general esterase during the sensitive period as a predictor of adult wing morph

To establish general esterase (GE) activity as a readout for physiological plasticity and a correlate of adult wing morph, we sampled hemolymph from 117 individuals that were developmentally halfway through the last juvenile stage (Day 5 at standard rearing conditions) and followed these individuals through to adulthood so that we could assign wing morph. This timepoint was chosen as it corresponds to peak differences in esterase activity between morphs in *G. assimilis* and *G. rubens* (Gu & Zera, 1994; Zera & Tiebel, 1989).

Hemolymph was collected in a pipette after snipping off the cercus of an individual with a pair of micro-scissors. Two microliters of hemolymph were diluted in 98 µl of a 0.1 M K phosphate buffer, pH 7.2, containing 0.1% β-mercaptoethanol. Samples were centrifuged at 14,000 g for 5 min at 4°C and then immediately used in the general esterase activity assay. All collections were performed in the mid-afternoon (14:00–16:00 Pacific Standard Time).

General esterase activity was quantified in hemolymph samples following Gu and Zera's (1994) modifications to the Woodring and Sparks (1987) method and adapted as a microplate assay. In a 96-well plate, 10 µl of the 50-fold diluted sample was added to 180 µl solution of 50 mM Mops [3-(*N*-morpholino)-propanesulfonic acid] and 10 µl of a 5 mM α-naphthyl acetate solution (dissolved in 95% ethanol) in triplicate for each sample. The mixture was incubated at 30°C for 60 min and then stopped with 33.2 µl of 0.3% Fast Blue Salt B and 3.5% sodium dodecyl sulfate (SDS) in water. After an initial blank reading without the SDS solution, absorbances were read at 605 nm and converted to concentration of α-naphthol using a linear standard curve made from known concentrations of the compound (0, 0.5, 1, 2.5, 5, and 10 mM). Concentrations were then converted to enzyme activity by dividing by the incubation period (60 min) to calculate enzyme activity (mM of α-naphthol/min). Pilot experiments were performed to determine the assay incubation times and hemolymph concentrations that resulted in a linear production of α-naphthol.

Testing plasticity in wing morph and general esterase activity in lines that vary in liability

Family crosses

We used a split-brood design with full siblings exposed to two seasonal rearing conditions (Cool SD and Warm LD) to quantify plasticity in morph frequency and general esterase activity in our polymorphic colony. We established 48 breeding pairs of 1-week-old virgin LW and SW adults from the polymorphic colony in a full factorial mating design, generating all four reciprocal crosses ($n = 12$ per cross: SW ♀×SW ♂, LW ♀×LW ♂, SW ♀×LW ♂, LW ♀×SW ♂). All pairs were housed in plastic containers (60 cm × 35 cm × 45 cm) under standard rearing conditions and were allowed to mate and oviposit for 15 days, with fresh oviposition substrate cups

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being provided every 3 days (5 total/breeding pair). All egg cups were housed in a container and allowed to hatch for a 2-week period. Of the 48 crosses, 33 produced sufficient numbers of offspring for good estimates of morph frequency in each of two seasonal conditions (>90 offspring surviving to adulthood per cross), resulting in the final sample sizes: SW × SW $n = 8$, LW × LW $n = 8$, SW ♀ ×LW ♂ $n = 7$, LW ♀×SW ♂ $n = 10$.

Juveniles from successful crosses, ranging from LW♀ × LW♂ and SW♀ × SW♂ to reciprocal mixed crosses (LW♀ × SW♂ and SW♀ × LW♂), were group-reared under standard conditions until the penultimate instar. Offspring from each family were then split between Warm LD and Cool SD conditions in a split-brood design, with separate cohorts used to measure adult wing-morph frequency ($n \approx 30$ siblings/family/treatment; $n = 1{,}854$ total) and general esterase (GE) activity ($n \approx 15$ siblings/family/treatment; $n = 836$ total). Individuals from each artificial-selection population were similarly reared under both environmental conditions. Because of the difference in development rates in the two environmental conditions, hemolymph was collected from individuals raised under the Warm LD condition at Day 5 of the ultimate juvenile stage, while individuals raised under the Cool SD condition were sampled at Day 9 of the ultimate juvenile stage. These days were determined from our prior experiments to be the middle of the last instar at those two temperatures (Table S1).

Artificial selection populations

We assessed wing-morph plasticity and GE activity plasticity in the artificially selected populations ($n = 6$; $n = 3$ LW-selected, $n = 3$ SW-selected). Penultimate juvenile crickets sampled from the eighth generation of selection ($n = \sim 270$/population) were randomly assigned to the Warm LD or Cool SD experimental treatments to complete development and phenotyped for adult wing morph ($n = \sim 90$/population/treatment; $n = 951$ total) and general esterase activity ($n = \sim 45$/population/treatment; $n = 405$ total) as described above.

Statistical analyses

All statistical analyses were conducted in R version 4.4.1. For each analysis, we fit a full model including all biologically relevant fixed effects and interactions, together with the appropriate random effects to account for the experimental design. For continuous response variables, distributions were assessed for approximate normality and homogeneity of variance prior to analysis. Model adequacy was evaluated by visual inspection of residual plots, Shapiro–Wilk tests of residual normality for Gaussian models, DHARMa residual diagnostics for generalized models, and assessments of convergence, singularity, and collinearity. Statistical significance of fixed effects in generalized (mixed) models was assessed using Type III ANOVA tests. Candidate models were compared only when evaluating alternative biological hypotheses (e.g., linear vs. quadratic relationships between liability and plasticity), with support assessed using Akaike's information criterion (AIC). Unless otherwise noted, values are reported as mean ± standard error of the mean.

To evaluate the response to selection in the artificial selection lines, we calculated the overall number of LW individuals of each sex in each generation (F1–F8) across the six replicate selection lines. We then fit a binomial generalized linear mixed-effects (GLMM) model with selection direction (LW-selected or SW-selected, polymorphic), sex (male or female), and generation as fixed effects, including all two-way and three-way interactions among fixed effects, and replicate population (P1, P2, P3) included as a random effect.

To test for environmental effects and determine the sensitive period for morph determination, morph frequencies within cohorts were analyzed using a binomial generalized linear model on individual-level data (LW = 1, SW = 0) with rearing condition (Warm LD, Cool SD, Transfer from Warm LD to Cool SD, Transfer from Cool SD to Warm LD) and sex (male or female) as fixed effects. To validate general esterase activity as a proxy for liability underlying morph determination, we applied a radical transformation to normalize the distribution of the data for analytic purposes. We then fit a linear model with both morph (SW or LW) and sex (male or female) as fixed effects, and their interactions.

For the family-level wing-morph frequency data, morph frequencies of siblings were analyzed using a binomial generalized linear mixed model with a probit link. The model included rearing environment (Warm LD or Cool SD), sex (male or female), and family cross type (SW × SW, LW × LW, or Mixed Cross [SW♀ × LW♂ and LW♀ × SW♂]) as fixed effects and their interactions, with family cross as a random effect. We first tested whether the two reciprocal mixed crosses differed in morph frequency; because they did not, they were combined into a single Mixed Cross category for subsequent analyses. A similar modeling approach was used for the artificial selection populations, with selection population (LW-selected or SW-selected) as a fixed effect and founding population (P1, P2, and P3) as a random effect.

For the family-level GE activity data, we removed outliers ($n = 3$) using the interquartile range method and again applied a radical transformation to better approximate a normal distribution. GE activities of siblings from individual crosses were analyzed using a generalized linear mixed model, which included rearing condition (Warm LD or Cool SD), sex (male or female), and cross-type (SW × SW, LW × LW, or Mixed Cross (SW ♀ ×LW ♂, LW ♀ × SW ♂) as fixed effects, and their interactions, and family as a random effect. The same modeling strategy was used for the artificial selection data, with cross replaced by selection population (LW or SW) and founding population (P1, P2, and P3) added as a random effect.

To test the hypothesis that phenotypic plasticity in wing morph depends on underlying liability relative to the threshold (Figure 1C), GE activity was standardized within each dataset (family crosses or artificial selection populations) by *Z*-score transformation prior to model fitting. Standardization was necessary because baseline GE activity differed between datasets, allowing effect sizes to be compared directly between the family crosses and artificially selected populations. We then fit a binomial generalized linear model for wing morph with genetic population (family crosses or artificial selection), rearing condition (Warm LD or Cool SD), and their interaction as fixed effects. A significant interaction term indicated genetic differences in phenotypic plasticity (i.e., G × E). An analogous Gaussian model was fit to standardized GE activity using the same fixed effects to test for corresponding G × E effects on the underlying physiology.

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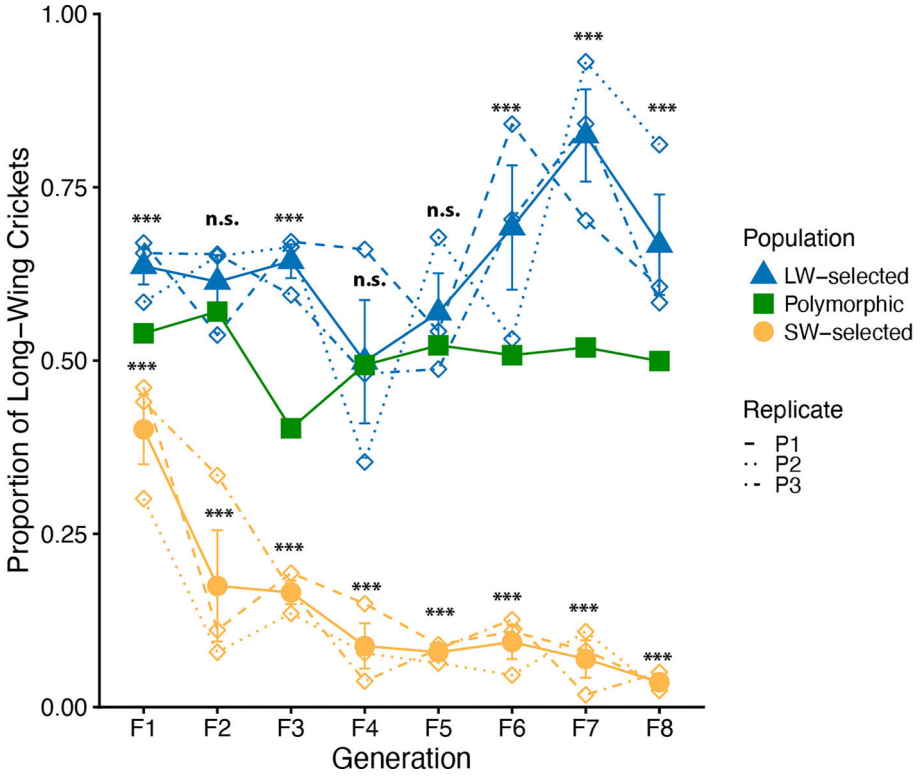


Figure 2. Divergence of morph frequencies of the replicate artificial selection populations (SW-selected, circle; LW-selected, triangle) in comparison to the existing polymorphic lab colony (square) across eight generations of selection. Filled symbols connected by solid lines denote the mean proportion of long-winged adults across the three replicates of the respective selection population, while open diamonds connected by differently styled lines represent individual replicates (P1, P2, P3: linetypes indicated in legend). Error bars represent standard errors of the mean proportion of long-winged adults across replicate selection populations (P1–P3) within each selection regime. Asterisks (***) denote a statistical difference ($p < 0.001$) in the mean morph frequency of the three replicate selection lines compared to the polymorphic colony within a generation, while (n.s.) denotes a non-significant difference.

To further evaluate the relationship between liability and plasticity, we estimated each family's or selection population's relative position along the liability distribution using its average long-wing morph frequency across both rearing environments, following the threshold-trait framework (Lynch & Walsh, 1998). Under this framework, the average probability of expressing the long-wing morph provides an empirical estimate of relative liability. We first tested the association between family- or population-level mean GE activity and this index of relative morph liability using a linear model. Wing-morph plasticity was calculated as the difference in long-wing morph frequency between Warm LD and Cool SD conditions within each family or selection population, and GE plasticity was calculated as the difference in standardized GE activity between environments. We then fit generalized linear models relating estimated liability to plasticity using either linear or linear-plus-quadratic terms for liability. Models were compared using AIC, with the better-supported model used to evaluate the predicted relationship between liability and plasticity (Figure 1C and Figure S9).

Results

Adult wing morph responds asymmetrically to selection

To generate extreme liabilities that are shifted away from the threshold in *G. lineaticeps*, we used an artificial selection experiment. Following prior work in *G. rubens* (Zera & Rankin, 1989) and *G. firmus* (Zera & Huang, 1999), we generated replicate artificial selection populations starting from a polymorphic lab colony of *G. lineaticeps*. After the eighth generation of selection, the average morph frequency of LW crickets across the three replicate SW-selected populations was 3.5% ± 0.76% LW, whereas across the three replicate LW-selected populations it was 66.7% ± 7.23% LW, compared to the polymorphic laboratory colony's average of ~50% LW (Figure 2). Despite the polymorphic colony being maintained at approximately equal proportions of LW and SW adults in each breeding cohort, we observed a slight but significant decline in the proportion of LW adults over the time frame corresponding to the first through eighth generations of the artificial selection experiment (from 54% to 50%; Fisher's exact test, $p < 0.01$, odds ratio = 1.17, 95% CI = 1.03–1.34).

The direction of selection (LW vs. SW), generation, and sex each had significant main effects on the proportion of LW adults with a significant three-way interaction (Table 1). This interaction reflects sex-specific differences in how morph frequencies changed across generations depending on the direction of selection. In SW-selected populations, both sexes rapidly converged on a low frequency of LW individuals by the first generation and remained consistently low thereafter. In contrast, LW-selected populations exhibited greater among-replicate variation and a more gradual, sex-dependent response to selection. Females showed a stronger response than males to selection on LW in early generations (F2–F4), but by the sixth generation, sex differences had

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Table 1. The effect of the direction of selection (for LW vs. for SW), sex, and generation on the proportion of LW adults in the artificial selection experiment (Figure 2).

Response variable	Explanatory variables	X^2	df1	df2	p
Proportion of LW adults	Direction of selection	**56.52**	**2**	**63**	**<0.001*****
	Generation	**98.86**	**7**	**63**	**<0.001*****
	Sex	**3.97**	**1**	**63**	**0.046***
	Direction of selection × generation	**448.52**	**14**	**63**	**<0.001*****
	Direction of selection × sex	1.60	2	63	0.449
	Generation × sex	**21.62**	**7**	**63**	**0.003****
	Direction of selection × generation × sex	**29.69**	**14**	**63**	**<0.001*****

Note. Type-III ANOVAs were performed. Significant effects are bolded. Asterisks denote significance levels($*p < 0.05$, $**p < 0.01$, $***p < 0.001$).

largely dissipated (Figure S1). Only after the sixth generation did morph frequencies in the LW-selected populations consistently differ from those of the polymorphic colony (Figure 2). Sex ratios within the SW morph remained approximately even across all populations and generations. In contrast, the sex ratio within the rare LW individuals produced in the SW-selected populations became female-biased in later generations (Figure S2). Together, these results indicate that the response to selection on wing morph was asymmetrical and that the tempo and magnitude of morph-frequency change depended jointly on selection direction, generation, and sex.

Establishing general esterase during the sensitive period as a predictor of adult wing morph

Here, we aimed to test whether general esterase (GE) activity during the sensitive period of morph determination is a good predictor of adult wing morph and thus a potential physiological indicator of the underlying liability. Shifting individuals during development between environmental conditions known to affect adult morph revealed that the sensitive period for morph determination in *G. lineaticeps* is sometime after the transition from late penultimate to early ultimate juvenile stage (Figure S3). When sampled at this stage of development, juveniles that became LW adults had higher GE activity (0.021 ± 0.0011 mM α-naphthol/min) than individuals that became SW adults (0.0093 ± 0.00070 mM α-naphthol/min) (LW vs. SW, $z = 6.834$, $p < 0.001$) (Figure 3). Within each morph, males and females did not differ in GE activity (LW female vs. LW male, $z = 1.307$, $p = 0.1$; SW female vs. SW male, $z = -1.075$, $p = 0.28$). Therefore, juvenile GE activity on Day 5 of the ultimate juvenile stage predicts the adult wing morph of an individual and provides a physiological indicator of the underlying liability in this species, as has been shown in other *Gryllus* species (Gu & Zera, 1994, 1996).

Testing plasticity in both wing morph and general esterase activity in lines that vary in liability

We tested the hypothesis that phenotypic plasticity in adult wing morph is dependent on the liability in *G. lineaticeps*, with the prediction that populations with extreme liabilities will not show morphological plasticity in response to rearing environment but may retain underlying physiological plasticity in GE activity. To test a range of liabilities, we assessed plasticity in both the artificial selection populations (representing putative extremes of liability relative to the threshold) and in families from our polymorphic colony (representing a distribution of liabilities that likely spanned the threshold).

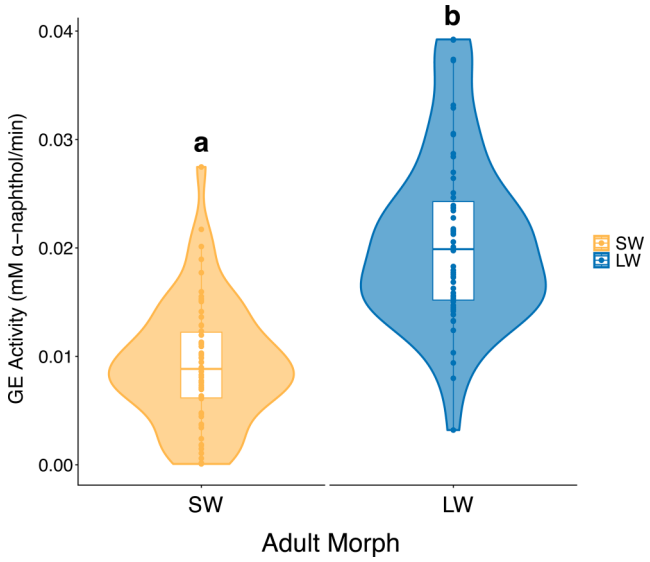


Figure 3. (a) General esterase (GE) activities of last instar crickets that were then phenotyped as either short-wing (SW, left) or long-wing (LW, right) upon reaching adulthood. Boxplots denote the 25th, median, and 75th quartiles, with whiskers denoting 1.5× the interquartile range, violin plots denote the probability density of the data, and solid dots denote individual crickets. Letters denote significant differences between groups ($p < 0.001$).

Variation in GE activity was explained by significant effects of family cross type, rearing environment, and a genotype-by-environment interaction, indicating genetic variation in plasticity (Figure 4A, Table 2). Individuals reared under Warm LD conditions had higher general esterase activity than those raised under Cool SD conditions (Table 2). Offspring from LW × LW families had the highest GE activity (0.034 ± 0.0014 mM α-naphthol/min), followed by mixed LW × SW families (0.032 ± 0.0009 mM α-naphthol/min) and SW × SW families (0.030 ± 0.0013 mM α-naphthol/min). While there was no main effect of sex on GE activity, sex did interact with family cross type and rearing environment (Table 2; Figure S4B). Visualization of the three-way interaction showed that males from SW × SW and LW × SW families exhibited a steeper increase in GE activity between Cool SD and Warm LD than did females, whereas males and females from LW × LW families increased similarly across environments (Figure S5).

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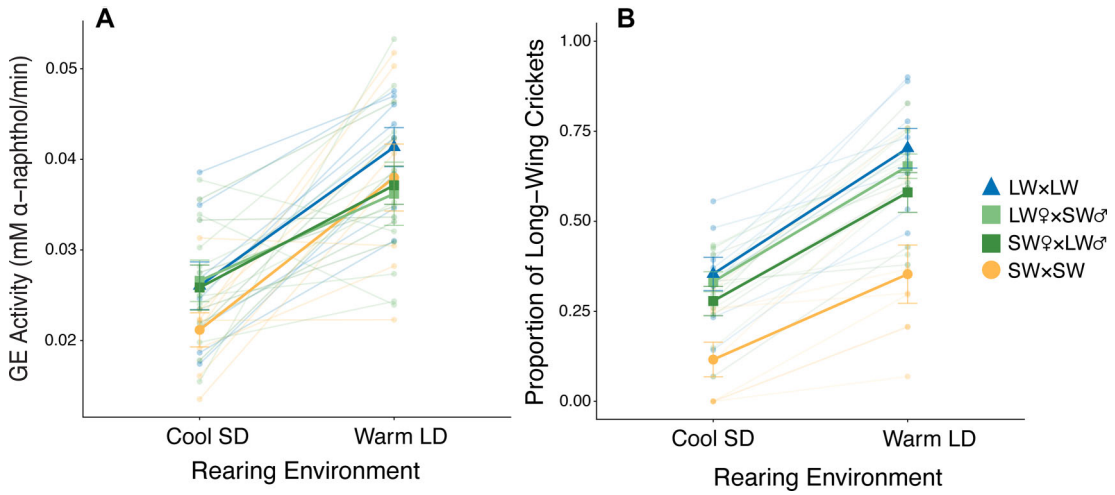


Figure 4. General esterase (GE) activity (A) and morph frequency (B) for *Gryllus lineaticeps* from split-brood family crosses (indicated by a color and symbol combination, see legend) raised under Cool Short Day (Cool SD: 24°C, 10 h:14 h Light: Dark) and Warm Long Day (Warm LD: 28°C, 16 h:8 h Light: Dark). Thin transparent lines represent individual families, while thick solid lines show the mean for each cross type. Error bars indicate the standard error of the mean.

Table 2. Tests of the effects of rearing environment, sex, and family cross type on GE activity in polymorphic crosses (Figure 4A) and artificial-selection crosses (Figure 5A).

Response variable	Explanatory variables	X^2	df1	df2	*p*
GE activity (polymorphic cross)	Rearing environment	**277.4**	**1**	**821**	**<0.001*****
	Sex	**1.4**	**1**	**821**	**0.237**
	Family cross	**10.56**	**2**	**821**	**0.005****
	Rearing environment × sex	**14.57**	**1**	**821**	**<0.001*****
	Rearing environment × family cross	**75.47**	**2**	**821**	**<0.001*****
	Sex × cross	**15.41**	**2**	**821**	**<0.001*****
	Rearing environment × sex × family cross	**24.79**	**2**	**821**	**<0.001*****
GE activity (AS population)	Rearing environment	**1160.92**	**1**	**395**	**<0.001*****
	Selection population	**879.26**	**1**	**395**	**<0.001*****
	Sex	**16.01**	**1**	**395**	**<0.001*****
	Rearing environment × selection population	**38.91**	**1**	**395**	**<0.001*****
	Rearing environment × sex	**3.99**	**1**	**395**	**0.046***
	Selection population × sex	0.25	1	395	0.616
	Rearing environment × sex × selection population	**3.84**	**1**	**395**	**0.05***

Note. Type III ANOVAs were performed. Significant effects are bolded. Asterisks denote significance levels($*p < 0.05$, $**p < 0.01$, $***p < 0.001$).

The pattern of environmental and genetic effects on GE activity was also evident in the artificially selected populations (Figure 5A, Table 2). GE activity differed between selection populations and rearing environments, and the significant population × environment interaction indicated genetic variation in physiological plasticity persists following artificial selection. Individuals from LW-selected lines had higher activity than those from SW-selected lines, and individuals reared under Warm LD conditions had higher activity than those reared under Cool SD conditions (Table 2; Figure 5A, Figure S6B). Sex also had a significant effect (Table 2), with significant interactions between rearing environment and sex, rearing environment and selection population, and a three-way interaction among all three factors (Table 2). Visualization of the three-way interaction demonstrated that within rearing environments and selection populations, mean GE activity was always higher in males than in females. Although LW-selected females showed a greater change in activity between rearing environments, males from SW-selected populations exhibited a greater overall increase in activity than females (Figure S7).

We next examined whether the physiological plasticity observed in GE activity was accompanied by plasticity in adult wing morph. In families derived from the polymorphic colony, morph frequency was significantly affected by both family type and rearing environment (Figure 4B; Table 3). Individuals reared under Warm LD conditions had higher LW proportions than those raised under Cool SD conditions (Table 3). LW × LW crosses produced the highest proportion of LW adults (0.53 ± 0.057), followed by mixed crosses (0.47 ± 0.033), and SW × SW crosses (0.23 ± 0.055). There were no significant interactions between any of the fixed ef-

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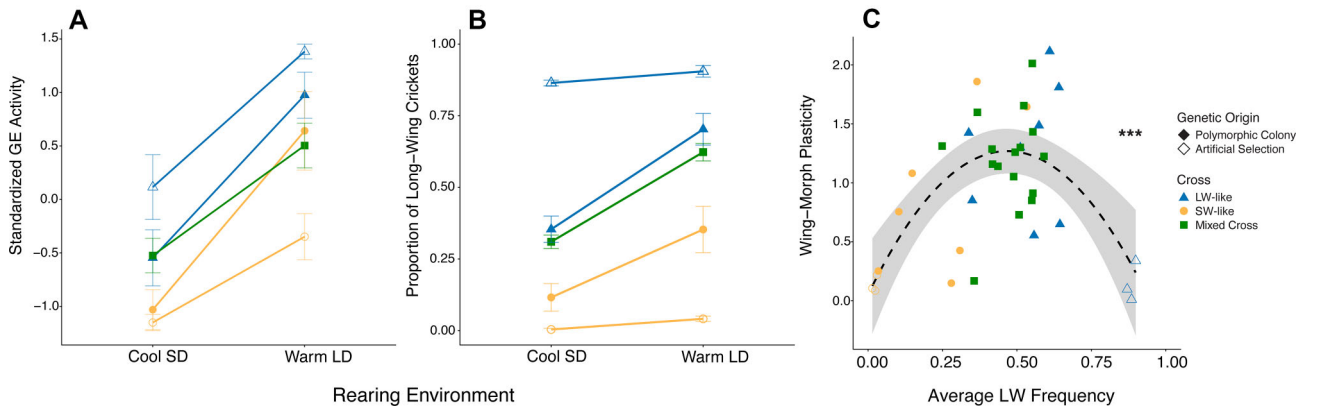


Figure 5. Linking physiological and phenotypic measures of liability to plasticity in *G. lineaticeps*. (A) Standardized (*Z*-score transformed) esterase activity and (B) wing-morph frequency for two genetic origins, the polymorphic colony (closed shapes) and artificial selection populations (open shapes), reared under two environmental conditions. Cross types are indicated by color and symbol (see legend). (C) Relationship between the average LW frequency and wing-morph plasticity. Average LW frequency was estimated as the average long-wing morph frequency of each family or population across both rearing environments, following the threshold-trait framework. Asterisks (***) denote statistical significance at $p < 0.001$. Error bars indicate standard error of the mean.

Table 3. Tests of the effects of rearing environment, sex, and family cross type on the proportion of LW adults in polymorphic crosses (Figure 4B) and artificial-selection crosses (Figure 5B).

Response variable	Explanatory variables	X^2	df1	df2	*p*
Proportion of LW adults (polymorphic cross)	Rearing environment	**19.7**	**1**	**1841**	**<0.001*****
	Sex	3.82	1	1841	0.051
	Family cross	**26.94**	**2**	**1841**	**<0.001*****
	Rearing environment × sex	1.54	1	1841	0.214
	Rearing environment × family cross	1.07	2	1841	0.585
	Sex × family cross	5.81	2	1841	0.055
	Rearing environment × sex × family cross	4.74	2	1841	0.093
Proportion of LW adults (AS population)	Rearing environment	2.46	1	942	0.117
	Selection population	**74.97**	**1**	**942**	**<0.001*****
	Sex	0	1	942	0.988
	Rearing environment × selection population	0.81	1	942	0.368
	Rearing environment × sex	0.81	1	942	0.368
	Selection population × sex	0	1	942	0.988
	Rearing environment × sex × selection population	0	1	942	0.988

Note. Type III ANOVAs were performed. Significant effects are bolded. Asterisks denote significance levels($*p < 0.05$, $**p < 0.01$, $***p < 0.001$).

fects (Table 3). Thus, unlike GE activity, which showed genetic variation in plasticity, the environmental response in wing-morph frequency was similar across families. In contrast to the environmental plasticity observed in general esterase activity after artificial selection, rearing environment had no significant effect on adult wing-morph frequency in the artificially selected populations, suggesting a loss of plasticity in wing-morph expression following selection (Table 3; Figure 5B, Figure S6A). There were also no significant differences in morph frequencies between the sexes and no interaction between selection population (G) and rearing environment (Table 3).

To further evaluate the relationship between GE activity and relative morph liability, we examined the association between family-level mean GE activity and mean LW frequency across rearing environments. GE activity was positively associated with mean LW frequency ($\beta = 13.92$, $t_{36} = 2.68$, $p = 0.011$, $R^2 = 0.17$; Figure S8), consistent with GE activity serving as a physiological correlate of relative morph liability.

To test our central hypothesis that phenotypic plasticity in wing morph is modulated by shifts in the underlying liability away from the threshold (Figure 1C), we combined the datasets from artificial selection lines, representing extreme liabilities, and family crosses from the polymorphic colony, representing liabilities spanning the threshold. The extent of environmental plasticity in morph frequency differed between the artificial selection populations and family crosses, as demonstrated by a significant rearing environment × population interaction (Table 4, Figure 5B). Poly-

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Table 4. Tests of the effect of rearing environment and genetic origin on plasticity in wing-morph frequency and GE activity in *G. lineaticeps*.

Response variable	Explanatory variables	X^2	df1	df2	*p*
Proportion of LW adults	Rearing environment	**179.51**	**1**	**74**	**<0.001*****
	Genetic origin	**40.11**	**1**	**74**	**<0.001*****
	Rearing environment × genetic origin	**52.71**	**1**	**74**	**<0.001*****
GE activity (scaled)	Rearing environment	**703.08**	**1**	**71**	**<0.001*****
	Genetic origin	**5.66**	**1**	**71**	**0.017***
	Rearing environment × genetic origin	**11.32**	**1**	**71**	**<0.001*****

Note. Type-III ANOVAs were performed. Significant effects are bolded. Asterisks denote significance levels($*p < 0.05$, $**p < 0.01$, $***p < 0.001$).

morphic families exhibited large environmentally induced shifts in morph frequency ($\Delta \approx 36\%$, $z = 7.54$, $p < 0.001$), whereas artificially selected populations showed no effect of the environment ($\Delta \approx +1\%$, $z = 0.115$, $p = 0.91$). Thus, as populations evolved to monomorphy, they also lost environmental plasticity in morph determination. In contrast, GE activity exhibited significant environmental plasticity in both the polymorphic families and artificially selected populations (Table 4, Figure 5A). The rearing environment × genetic origin interaction was significant ($\chi^2 = 11.32$, $p < 0.001$), indicating that the magnitude of physiological plasticity differed between genetic origins. GE activity increased significantly between rearing environments in both polymorphic families ($\Delta = +1.74$ SD, $z = 7.28$, $p < 0.001$) and artificially selected populations ($\Delta = +1.14$ SD, $z = 16.35$, $p < 0.001$). Thus, artificial selection reduced phenotypic plasticity at the level of wing morph while preserving environmental sensitivity of the underlying physiology.

To test whether wing-morph plasticity had a unimodal relationship with liability (Figure 1C), we estimated the relative liability of each family or population as its mean proportion of LW adults across the two rearing environments, following the threshold-trait framework (Lynch & Walsh, 1998). Phenotypic plasticity was quantified as the difference in LW morph frequency between the Cool SD and Warm LD environments. The quadratic model was strongly favored over the linear model (ΔAIC = 18.63; likelihood-ratio test: $\chi^2 = 20.63$, $p < 0.001$) and explained 44.5% of the variation in wing-morph plasticity ($R^2 = 0.445$; adjusted $R^2 = 0.396$; Figure 5C; Figure S9).

Discussion

Selection on threshold traits decouples phenotypic and physiological plasticity

A major challenge in studying threshold traits is that variation in the underlying liability that does not span the critical threshold produces a single observable phenotype, thereby rendering trait variation due to genetic and environmental factors latent. This mapping between a continuous liability and discrete phenotypes complicates efforts to quantify plasticity, predict evolutionary dynamics, and link genotype to phenotype for threshold traits (Reid & Acker, 2022). Some polymorphic traits are entirely genetically controlled (e.g., coloration in rock pocket mice and some wing-dimorphic aphids) (Braendle et al., 2006; Nachman et al., 2003), while others are environmentally regulated (e.g., seasonal morphs in butterflies and temperature-dependent sex determination in reptiles) (Brakefield & Larsen, 1984; Holleley et al., 2015). However, many threshold traits have both genetic and environmental influences (G × E) (Hazel, 2002; Tuttle et al., 2016) such as the adult wing morph in *G. lineaticeps* we investigated here. Because the underlying liability of threshold traits is unobservable, it is difficult to determine whether evolutionary changes arise from shifts in genetic liability, changes in environmental responsiveness, or both. To address this question, we used controlled crosses and artificially selected populations of *G. lineaticeps* to generate a broad range of genetic liabilities while manipulating ecologically relevant environmental conditions. Plasticity was quantified for both the discrete adult wing morph and general esterase activity, which we established as a continuously variable physiological predictor of wing morph associated with underlying liability (Figure S8).

Our study provides empirical support for the prediction that phenotypic plasticity in dimorphic threshold traits is a function of the underlying liability and becomes latent as liability moves farther away from the threshold. In our crosses using crickets derived from a polymorphic population, we observed substantial plasticity in both wing morph and general esterase activity in response to seasonally relevant environmental variation. In contrast, in the artificially selected populations derived from the polymorphic population, phenotypic plasticity in wing-morph frequencies was nearly absent, while plasticity in general esterase activity remained (Figure 5A, B). We infer that individuals from polymorphic laboratory colony have liabilities in close proximity to the threshold, allowing environmental inputs to induce crossing of the threshold and result in observable wing-morph plasticity. Conversely, we infer that artificial selection populations shifted their liability distributions farther from the threshold, resulting in loss of wing-morph plasticity, despite retaining physiological plasticity (Figure 5C).

Studies have demonstrated that directional selection on a trait can reduce phenotypic plasticity in that trait, leading to populations or genotypes that are fixed for the phenotype even in the presence of genetic or environmental variation (Levis & Ragsdale, 2025). The term genetic assimilation describes the evolution of a constitutively expressed phenotype from an ancestrally environmentally sensitive developmental system (Nijhout et al., 2021), while traits that are invariant in the presence of genetic and/or environmental variation are referred to as canalized (Emlen, 1996; Waddington, 1953). Similarly, selection for plasticity can increase phenotypic responses to environmental variation, even in an initially monomorphic population (Grenier et al., 2016; Levis, 2026; Suzuki & Nijhout, 2006). In threshold traits, canalization can result when selection moves liabilities further from the threshold, reducing the probability that environmental signals or genetic variation produces phenotypic change (Suzuki & Nijhout, 2006). Our results are consistent with

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this model, as selection on adult wing morph greatly reduced environmentally induced morph variation, while general esterase activity, a physiological trait associated with liability, remained sensitive to the environment (Figure 5). We note, however, that our artificial selection could have changed the location of the threshold with respect to the liability (Fairbairn & Yadlowski, 1997), rather than shifting liability values relative to the threshold. Furthermore, the reduction in plasticity observed under artificial selection may not occur under natural selection, where weaker or opposing selective pressures could produce different evolutionary outcomes (Levis & Ragsdale, 2025).

Asymmetric evolutionary trajectories under divergent selection

Artificial selection provides a powerful approach for evaluating how traits respond to directional selection (Garland & Rose, 2009; Hill & Caballero, 1992). In addition to the loss of wing morph plasticity, we observed an asymmetric response to selection for the two different wing morphs: Selection toward the SW morph produced a rapid and consistent response across the replicate populations, whereas selection toward the LW morph resulted in a slower, more variable response. Assuming that this asymmetry is not a consequence of rearing crickets in a laboratory, it suggests that the evolutionary loss of dispersal capability may proceed more readily than its gain, which is further supported by the predominance of short-wing monomorphism in the North American *Gryllus* clade (Treidel et al., 2026). Notably, even the polymorphic laboratory colony exhibited a modest, yet significant decline in LW morph frequency over time, despite breeding individuals being selected at equal morph ratios. These observations may help explain patterns of wing-morph evolution across *Gryllus*, where the inferred instances of the evolution of monomorphic short-wing species are much more common than evolution to monomorphic long-wing species (Treidel et al., 2026). Our findings suggest that the evolution of wing monomorphy may involve changes in the developmental regulation underlying wing-morph plasticity, rather than simply the loss of one morph. Although differences in the trade-offs between morphs have been traditionally proposed as important drivers of wing-polymorphism, recent work suggests that these trade-offs may be less pronounced than previously thought, at least in females (Laturney et al., 2026). The mechanisms underlying the apparent asymmetry in the evolution of wing monomorphy therefore remain an important question for future research.

Differences in wing morphology between males and females of the same species are commonly observed in wing-polymorphic insects, reflecting sex-specific dispersal strategies (Caillaud et al., 2002; Martínez-Pérez et al., 2022). In our LW-selected populations, females initially responded more strongly than males (Figure S1), whereas in SW-selected populations, the rare LW individuals that appeared in later generations were predominantly female (Figure S2). In the crosses derived from the polymorphic colony, males showed greater plasticity in esterase activity than females (Figure S5), but in the artificial selection populations, this difference was diminished and inconsistent between the LW- and SW-selected populations. This suggests that sex differences in the expression and physiology of wing polymorphism are context dependent and can evolve rapidly under selection.

Seasonal plasticity in wing dimorphism and its underlying physiology

Environmental variation during early juvenile life-stages can serve as a cue for life-history strategies expressed in adulthood (Gershman et al., 2022; Min et al., 2021; Santos et al., 2021). When we transferred *G. lineaticeps* to a different seasonal environment between the penultimate and ultimate juvenile stage, we observed wing-morph frequencies consistent with the second environment, suggesting that these individuals were within the sensitive period for morph determination. Our results are consistent with findings in other *Gryllus* species in which the sensitive period is a late juvenile life-stage (Roff & Shannon, 1993; Zera & Tiebel, 1989; Zhao & Zera, 2006). Prior studies of *Gryllus* have reported seasonal fluctuations in morph frequency, with summer-like conditions correlating with a greater proportion of LW adults and spring or autumn-like conditions correlating with a greater proportion of SW adults (Roff et al., 2002). Consistent with this, in our experiments with *G. lineaticeps*, summer-like conditions promoted more LW adults, and cooler, shorter day, conditions promoted more SW adults.

While seasonal plasticity in wing-morph determination is evident among many wing dimorphic species (Brisson, 2010; Zera & Denno, 1997), our understanding of the complete causal mechanism is incomplete (Zera, 2003; Zhang et al., 2019). We found that general esterase (GE) activity in juveniles was positively correlated with the likelihood of becoming a LW adult and showed for the first time that GE activity is plastic in response to environmental variation. Summer-like conditions increased GE activity during the period of morph determination and increased the proportion of LW adults. The GE assay provided a non-specific measure of ester hydrolyzing activity that may reflect the activity of carboxylesterase enzymes, a large family of enzymes involved in chemical detoxification, hormone and pheromone metabolism, and neuronal development (Oakeshott et al., 2005). Prior work employing the same assay demonstrated tight phenotypic and genetic correlations between GE activity and specific activity of JHE across juvenile development of *Gryllus*, suggesting these two traits co-vary within individuals and should evolve together (Gu & Zera, 1994, 1996). Since JHE acts as the primary regulator of JH titers during the critical period of morph determination in *Gryllus* (Fairbairn & Yadlowski, 1997; Zera & Cisper, 2001; Zera & Denno, 1997; Zera & Harshman, 2001; Zera & Huang, 1999; Zera & Tiebel, 1989; Zhao & Zera, 2006), we propose a model where seasonal upregulation of GE or JHE expression leads to increases in LW morphs. Other hormones or signaling molecules (e.g., ecdysone, insulin-like peptides) likely work in concert with JH (Lin & Lavine, 2018), and additional endocrine and cell signaling pathways likely modulate downstream physiological processes to ultimately influence morph determination (Zera & Huang, 1999). Elucidating the relationship between GE activity and its downstream consequences for endocrine signaling and physiology, as well as understanding what aspects of *G. lineaticeps* ecology have shaped the evolution of seasonal plasticity in these pathways, remain important areas of study in this system.

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In conclusion, we used the threshold trait model to interpret how direct selection on wing morph impacts phenotypic plasticity in *G. lineaticeps*. Our results indicate that the evolution of more extreme liabilities may lead to a loss of adult morph plasticity even while the underlying trait liability remains sensitive to the environment. Consequently, a population or species observed to be monomorphic for wing length may nonetheless harbor latent plasticity that could allow for the appearance of wing polymorphism under extreme environmental exposures or following changes in allele frequencies at loci underlying the distribution of liability.

Supplementary material

Supplementary material is available online at *Evolution*.

Data availability

Data are available from Dryad: https://doi.org/10.5061/dryad.jsxksn0qn.

Author contributions

L.M.A.M., J.E.L., L.A.T., C.D.M., K.L.M., and C.M.W. designed the research. L.M.A.M., L.J.D., O.M.R., J.E.L., and A.G.K. performed the research. L.M.A.M. analyzed the data. L.M.A.M., J.E.L., L.A.T., K.L.M., and C.D.M. wrote the paper. C.D.M. and K.L.M. acquired funding. C.M.W. acquired funding and provided new reagents and analytical tools.

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Conflict of interest

The authors declare no conflict of interest.

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